Advanced CFD Homework 1

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Contents

1 Introduction

This report investigates the performance of two different meshing approaches, morphing and overset meshing. The process of this assignment is shown in [1] and [2] along with the flow case considered; flow past two circular cylinders.

The upstream cylinder is attached to a spring with spring constant $k = 1.027$ [N/m] causing oscillatory vertical motion. As described in [1], the spring constant was chosen such that the dimensionless parameters U^* cause the vacuum frequency to coincide with the vortex shedding frequency. The natural (vacuum) frequency of the oscillating system was found to be (Figure 1) using the period of oscillations for the lift force on a fixed upstream cylinder.

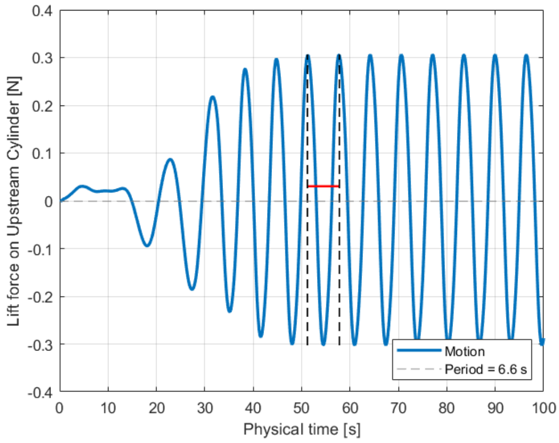


Figure 1: Lift force [N] on fixed upstream cylinder with time [s]

2 Part 1 - Morpher

Morphing is a technique of dynamically altering the shape and size of a mesh to adjust to changes in the geometry of moving or deforming objects. This process utilizes control points, which define how the mesh will change and deform (thinning) during the simulation, allowing for real-time updates to the mesh before interpolation takes place [3].

2.1 Why not rigid translation of the mesh?

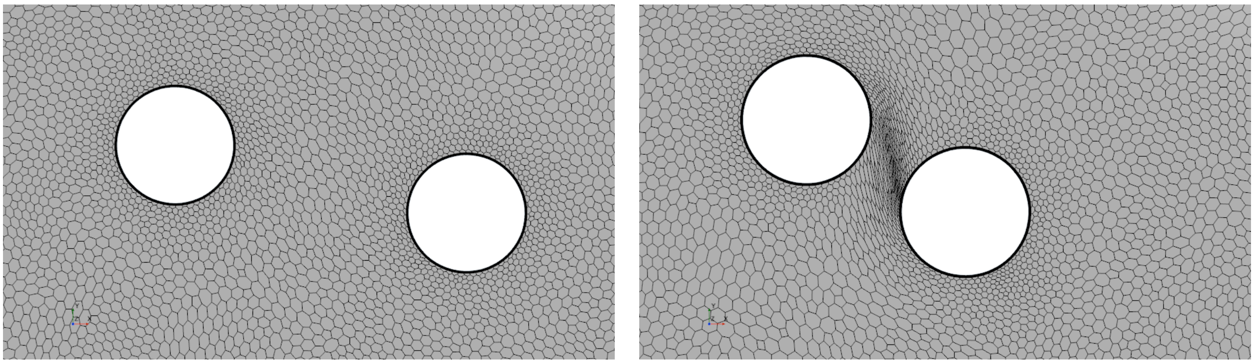
Rigid translation of the mesh is an inadequate choice due to the complexity of the fluid-structure interaction (FSI) present. The flow field around a circular cylinder is highly unsteady, causing vortex shedding which leads to a vortex-induced vibrations (VIV) motion.

A rigid translation would move the whole mesh as one entity, without any deformation or local adaptation. Therefore certain necessary flow changes, like local velocity fluctuations and pressure gradients close to the oscillating cylinder would not be captured.

Furthermore, the expected motion from VIV is a non-linear (oscillatory) motion. Coupling the FSI in this case requires the mesh to deform locally so that the interaction of the cylinder with the surrounding vortices can be accurately resolved.

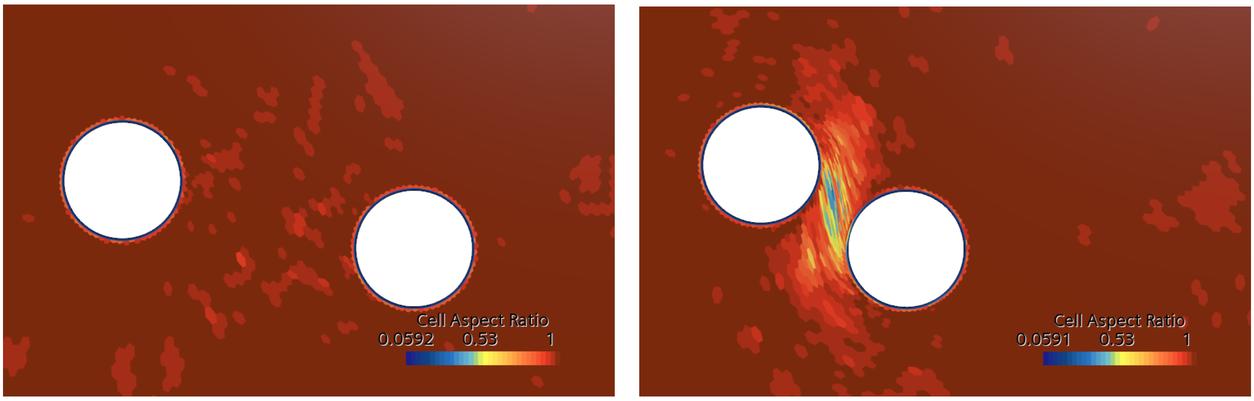
2.2 Performance of the morpher for spacing $1.5D$ and $0.25D$.

The performance of the morpher for flow cases with cylinder spacing 1.5D and 0.25D with identical physics is analyzed by visualizing the mesh (Figure 2). The reduced distance



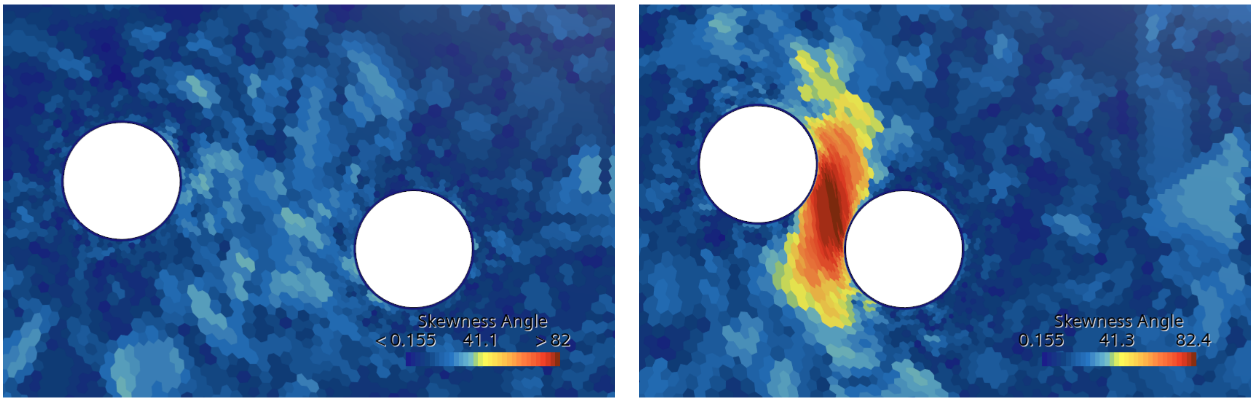
(a) 1.5D (b) 0.25D

Figure 2: Scalar field of flow case at peak of oscillation showing the Morpher mesh



(a) 1.5D (b) 0.25D

Figure 3: Scalar field of flow case at peak of oscillation showing the aspect ratio of the Morpher mesh



(a) 1.5D (b) 0.25D

Figure 4: Scalar field of flow case at the peak of oscillation showing the skewness angle of the Morpher mesh

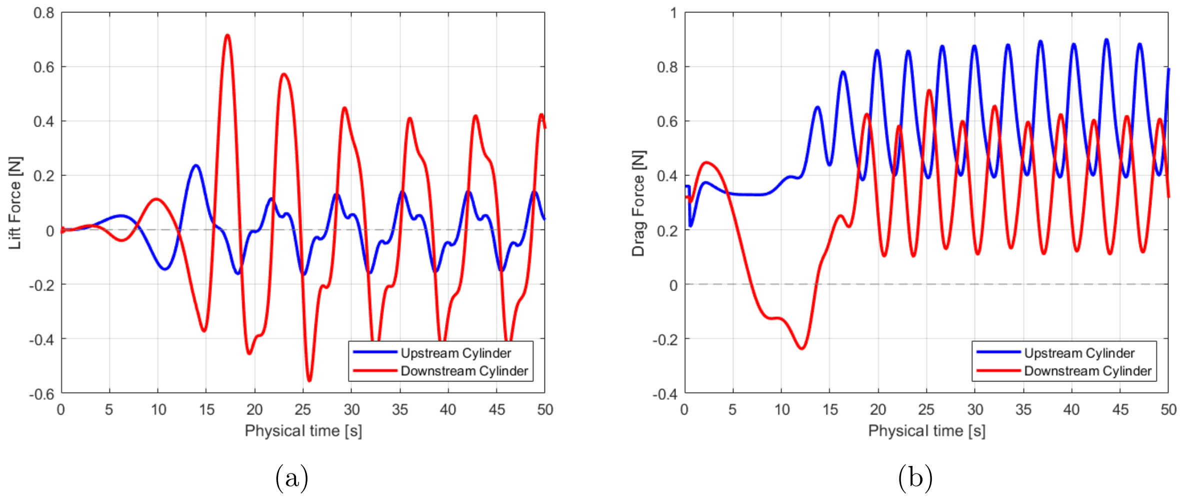


Figure 5: Aerodynamics forces (a) Lift, (b) Drag [N] exerted on both cylinders for 1.5D spacing using morpher

between the components in the simulation, particularly in the 0.25D test, resulted in a significantly higher mesh distortion (Figure 2b). The effects of this distortion on the quality of the cells are evident when looking at the cell aspect ratio (Figure 3) and skewness angle (Figure 4), as suggested by SIEMENS user manual [4].

Both metrics report the amount of distortion in the mesh and have agreed-upon limits to determine if the cell quality is acceptable. Cells with a skewness angle greater than 85° are considered "bad" cells [4]. A skewness angle of lower than 41.1° is seen for the 1.5D case (Figure 4a), however, in the 0.25D case (Figure 4b), the cells reached a skewness angle of 82.4°. While not exceeding the threshold, this skewness may still result in numerical errors or inaccuracies. The same can be said about the cell aspect ratio for the 0.25D case (Figure 3b) which reports distorted cells.

This suggests that the morpher can handle simpler geometries such as the 1.5D spacing but may not resolve the flow adequately enough in more complex and tight geometries such as 0.25D spacing. The impact of this inaccuracy is difficult to judge based on the mesh distortion alone, yet should be taken into account when concluding.

2.3 Aerodynamics forces on cylinders using morpher

This flow case is expected to experience VIV, where the upstream cylinder will oscillate upwards and downwards (y-direction). Such a motion can be observed in the three snapshots (Figures 14,15,16) of the velocity vector field, which illustrate the flow as shown in the Appendix 5. Patterns seen in the velocity vortex field can be related to the aerodynamics force lift (Figure 5a) and drag 5b).

Lift is generated due to pressure differences between the upper and lower surfaces of the cylinders [5], which are driven by how the velocity changes across the surface. A bluff object such as a cylinder may cause vortex shedding (von Kármán vortex street), which alternates in shedding and leads to fluctuating lift known as VIV [6]. As VIV occurs on the upstream cylinder, which is a region of low pressure, a wake is formed in front of the downstream cylinder. This wake has a non-uniform velocity and causes larger pressure differences leading to a larger lift force.

Drag is caused by two reasons. The pressure differences between the front and back side of the cylinders, and skin friction drag which is equal in magnitude for both cylinders and can be considered negligible for a bluff body [5]. Furthermore, a wake is formed when the flow separates from the object's surface creating a low-pressure region which contributes to the drag force. This separation can be observed in the flow through an adverse pressure gradient or recirculating velocity vector. The downstream cylinder can therefore be said to experience less drag because the incoming fluid has a lower pressure due to the wake of the upstream cylinder.

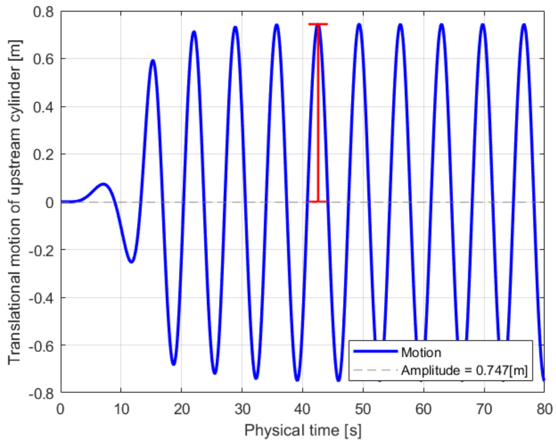


Figure 6: Translational motion of upstream Cylinder [m] with time [s], showing amplitude of motion is 0.747 [m]

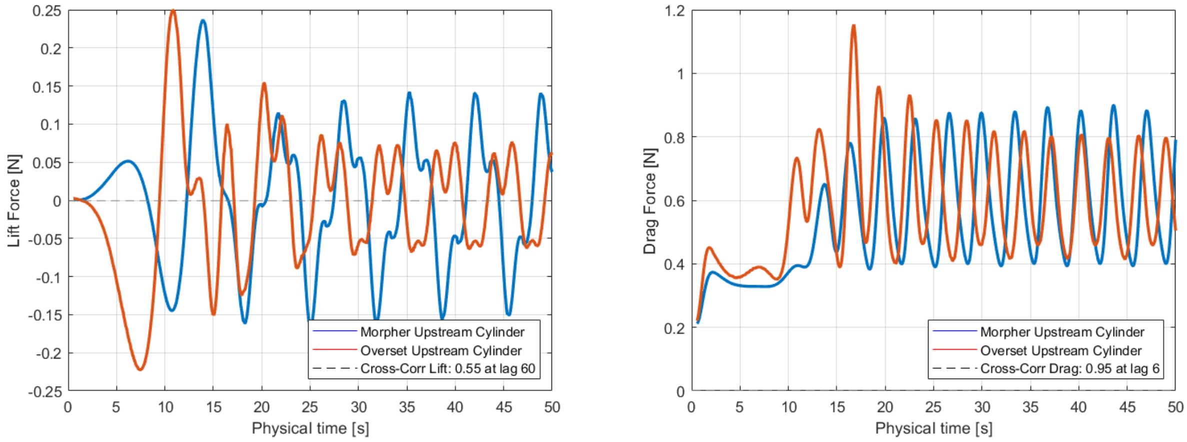
It is also important to note that the forces experienced before the system reaches a steady state (approximately at $t = 30$ [s]) largely depend on how the simulation numerically initiates the vortex shedding and wake formation. For instance, the first vortex could be formed above the cylinder or below. Such transient effects depend on the numerical perturbations used.

2.4 Cylinder motion

The translational motion of the upstream cylinder (Figure 6) reaches a steady state after approximately 40 [s]. After this time the amplitude no longer changes and is found to be 0.747 meters.

3 Part 2 - Overset

Overset meshing uses multiple (in this case two) overlapping meshes to simulate complex geometries and moving objects. Each mesh works independently, allowing for DFBI motion without needing to re-mesh the whole domain. Interpolation then takes place simultaneously between overlapping regions, to ensure accurate flow variable transfer [7].



(a) (b)

Figure 7: Aerodynamics forces (a) Lift, (b) Drag [N] exerted on both cylinders with cross-correlation for 1.5D spacing using morpher and overset

3.1 Comparison of mesh deformation techniques

Differences between the morpher and overset meshing can be seen when looking at the aerodynamic forces exerted on the upstream cylinder (Figure 7). An apparent difference is the amplitude and direction of the first peak of the lift force (Figure 7a) which is not only significantly larger for the overset mesh but also in the opposite direction. This difference happens because of the way with which vortex shedding is initiated, specifically the numerical perturbation that the two methods use. Once the boundary layers have formed around the cylinders and the shedding pattern causing the interaction between the cylinders has developed, the methods seem to resemble a somewhat similar shaped pattern. This is supported by the cross-correlation of 0.55 (0 = no correlation, 1 = strong correlation) with a lag of 60 [s] (time shift). On the other hand, the drag force (Figure 7b) is seen to show a much stronger correlation with a cross-correlation of 0.95 and lag of 6 [s].

Another difference, which is more apparent when comparing the translational motion (Figure 8), is that the overset mesh experiences a decaying motion after reaching a steady state, whereas the morpher does not. This leads to a rather poor cross-correlation of 0.10, which may result from the mesh quality of the two methods. As the cylinder moves through the fluid it experiences viscous damping causing the oscillations to decay. The morpher may not be able to resolve the flow on this scale because of interpolation errors and neglecting the dissipation terms. Alternatively, the vortex shedding might not match the natural frequency of the system perfectly, which may have effects on the translation.

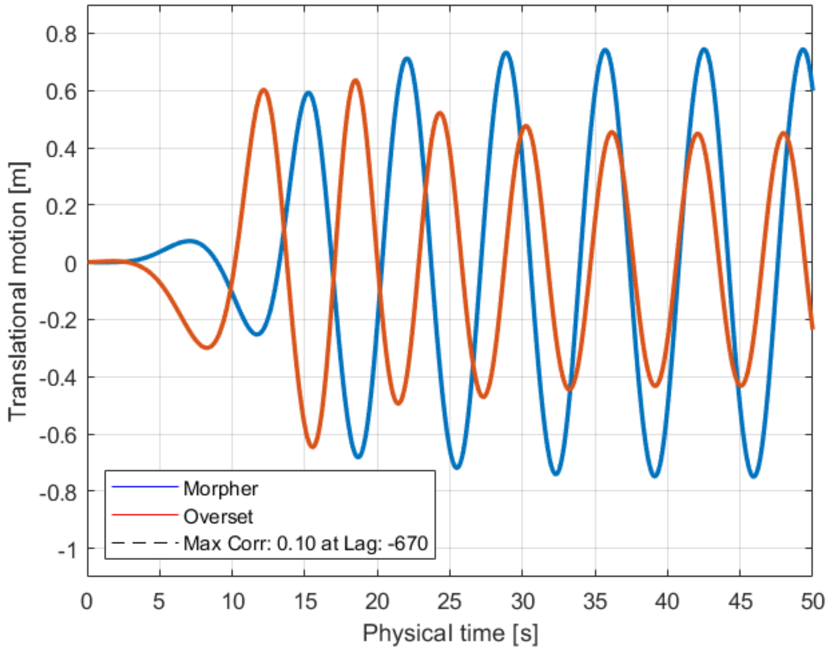


Figure 8: Translational motion for mesh deformation techniques, morpher and overset meshing, with cross correlation

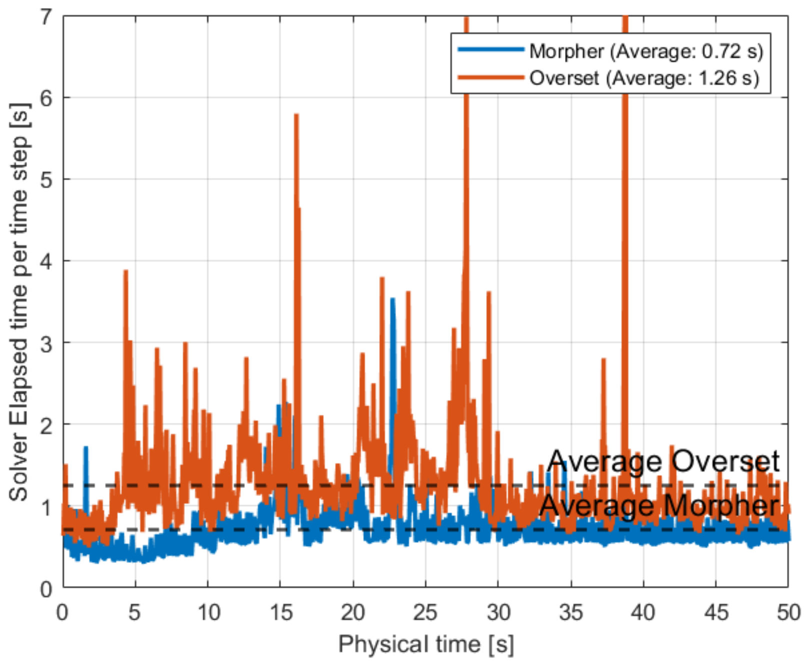
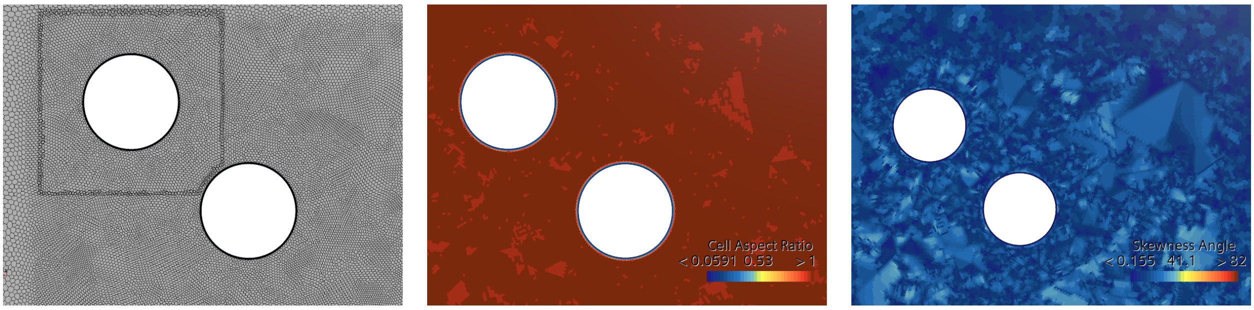


Figure 9: Solver elapsed time for mesh deformation techniques, morpher and overset meshing, with cross correlation



(a) Mesh (b) Cell aspect ratio (c) Skewness angle

Figure 10: Scalar field of the 0.25D flow case at the peak of oscillation using overset mesh

Solver elapsed time per time step indicates the amount of time it took the simulation to compute all necessary aspects such as discretizing and solving the equations for that time step [8]. Comparing it to the two simulations (Figure 9) indicates the performance of the different mesh deformations. On average the overset mesh took 1.26 [s] to compute a time step, whereas the morpher spent 0.72 [s]. The overset mesh's longer computation time is due to it having to generate two meshes instead of one.

3.2 Can overset meshing simulate the flow for spacing of 0.25D?

Separating the mesh into two regions and partially overlapping them in the interpolation region (Figure 10a), as done by the overset, is optimal for complex geometries exhibiting movement for the following reasons [7].

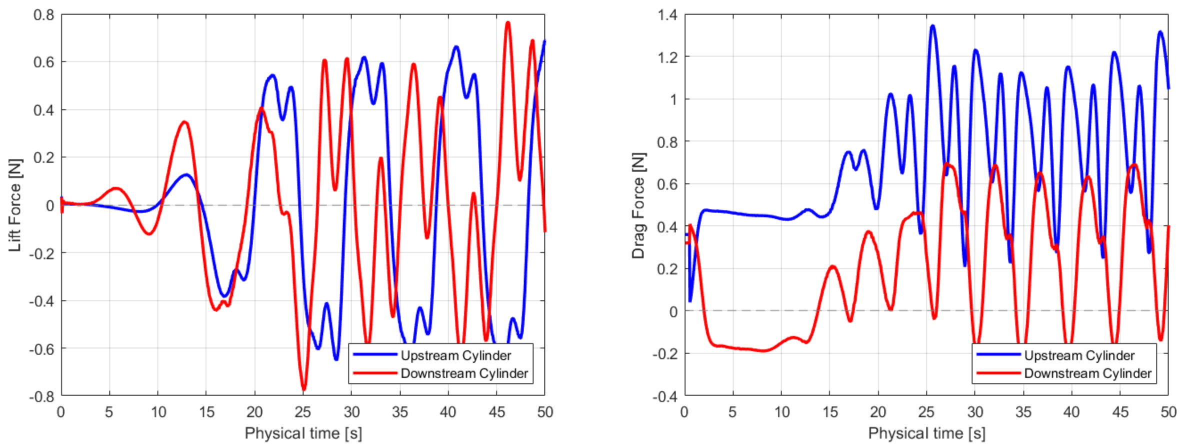
Firstly, the overset prevents the mesh from distorting by allowing it to move independently. This option is preferred over the morpher method because it is less likely to result in solver instabilities. When cells are distorted the approximations when solving may resolve the flow poorly or cause errors which are then amplified. Such can be seen through the cell aspect ratio (Figure 10b) and skewness angle (Figure 10c).

Secondly, the overset method implements the hole-cutting process, that checks if a boundary face resides inside or outside of the other overset region and adjusts the mesh accordingly [7]. For these reasons, the overset mesh can simulate the flow for spacing of 0.25D and give back much more relevant results.

3.3 Aerodynamics forces on 0.25D spaced cylinders using overset

The observed flow shown for reference in Appendix 5 illustrates a similar behaviour of the flow as in the case of the 1.5D space cylinders with a few differences in the exerted forces (Figure 11).

With a closer spacing, the downstream cylinder is located directly in the wake of the upstream cylinder. This lower pressure region and vortices shed by the upstream cylinder create an upwash which leads to more lift (Figure 11a), and less drag (Figure 11b) exerted on the downstream cylinder. This is the same reason why birds fly in a V-formation, as they can conserve their energy due to the favourable flow conditions [5]. Due to this flow between the two cylinders being highly constrained, a larger lift force can also be seen on the upstream cylinder, which may result from a different vortex shedding frequency compared to 1.5D.



(a) (b)

Figure 11: Aerodynamics forces (a) Lift, (b) Drag [N] exerted on both cylinders for 0.25D spacing using overset

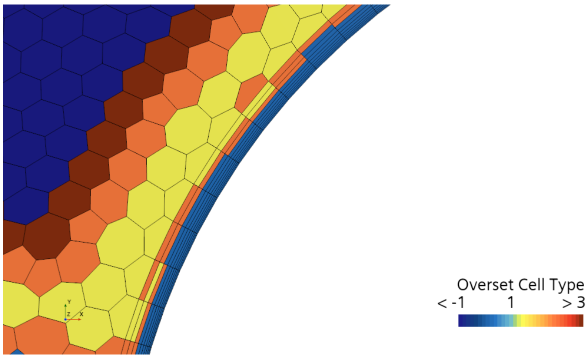
3.4 Impact of interface interpolation on acceptor-donor dependency

To connect the background with the overset region, a two-step process takes place; hole-cutting, which determines which cells are active (= 0), inactive (= -1), or acceptor cells (= 3), and donor Search, which ensures that donor cells (= 1) are found for each acceptor cell [7]. The resulting cell type after this process is then visualized using the acceptor-donor dependency.

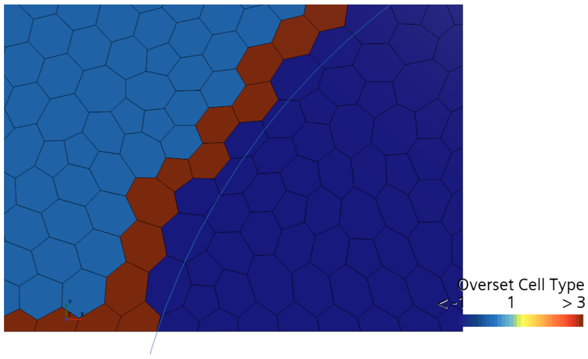
The acceptor-donor dependency for the linear (Figure 12) and least square (Figure 13) interpolation show minimal to no difference[1]. In the examined regions the overset region (Figure 12b,13b) has no difference and the background region (Figure 12a,13b) differs by very few acceptor cells closer being closer to the cylinder.

These findings suggest that the interpolation has almost no influence on the results, suggesting that the model is robust. Therefore there is sufficient resolution in the mesh to provide an adequate result.

[1]At the time of which this report was made, inconsistency in reporting the doner cells by Simcenter STAR-CCM+ has been reported [9], and has been therefore left out of this report.

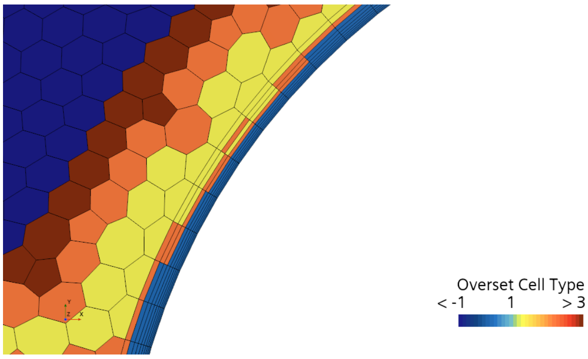


(a) Background region

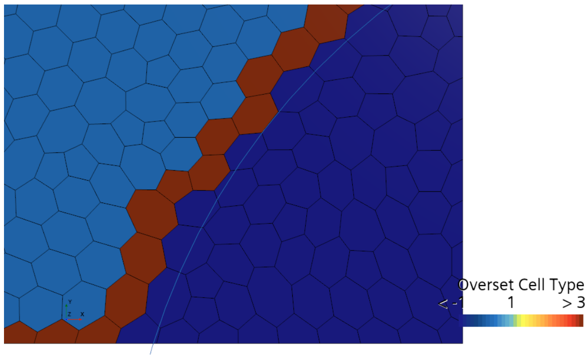


(b) Overset region

Figure 12: Acceptor-donor dependency for linear interpolation



(a) Background region



(b) Overset region

Figure 13: Acceptor-donor dependency for least squares interpolation

4 Conclusion

When comparing the morpher and overset meshing for the two flow cases, it can be stated that morphing is an effective technique for simple geometries with moderate movement because it yields similar results to the overset for the 1.5D flow case, while having a faster computing time. However, for more complex geometries and movements, especially when components might be close or even in contact with each other, implementing an overset of independent meshes provides the benefits of better accuracy and fewer numerical instabilities caused by mesh distortion.

List of Figures

Nomenclature

DBFI Dynamic Fluid Body Interaction

FSI Fluid Structure Interaction

VIV Vortex induced vibrations

References

[1] J. H. Walther, "ADVANCED CFD HOMEWORK #1, PART 1/2 VORTEX INDUCED VIBRATION â MORPHER," Tech. Rep. 1, DTU.

[2] J. H. Walther, "ADVANCED CFD HOMEWORK #1, PART 2/2 VORTEX INDUCED VIBRATION â OVERSET," Tech. Rep. 1, DTU.

[3] J. H. Walther, "ADVANCED CFD - Morphing," June 2024.

[4] *Simcenter STAR-CCM+ 2406 User Guide, Mesh Quality.*

[5] J. D. Anderson, *Fundamentals of aerodynamics.* McGraw-Hill series in aeronautical and aerospace engineering, New York: McGraw-Hill, 2nd ed ed., 1991.

[6] "Karman Vortex Street - an overview | ScienceDirect Topics."

[7] J. H. Walther, "ADVANCED CFD 41316 - Overset Meshing," Sept. 2024.

[8] J. H. Ferziger and M. PericĚ, *Computational methods for fluid dynamics.* Berlin ; New York: Springer, 2nd rev. ed ed., 1999.

[9] J. H. Walther, "overset cell type," Sept. 2024.

5 Appendix A

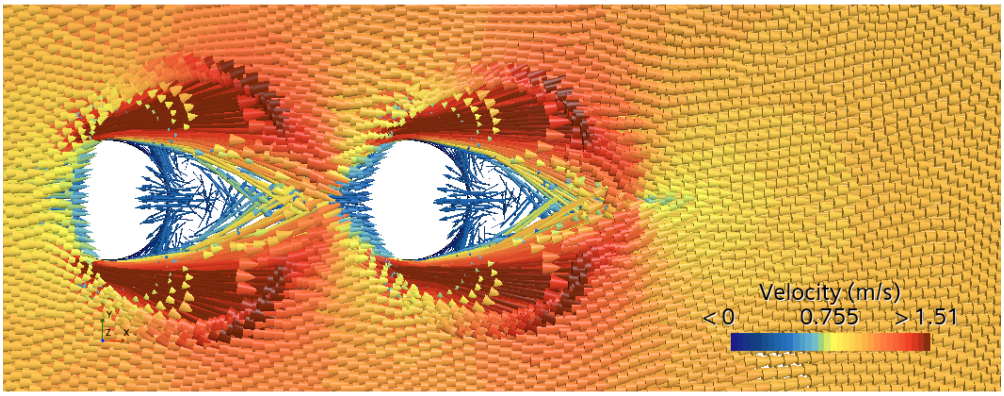


Figure 14: Velocity field for 1.5D cylinder using morpher at time = 0 [s]

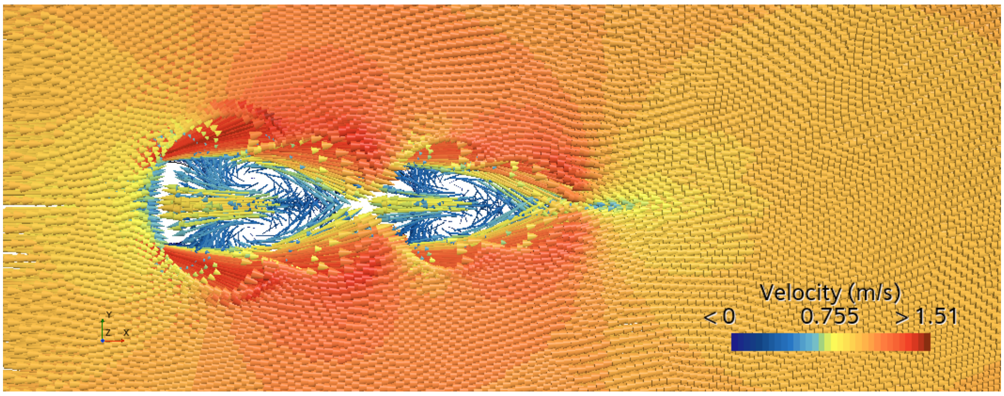


Figure 15: Velocity field for 1.5D cylinder using morpher at time = 5 [s]

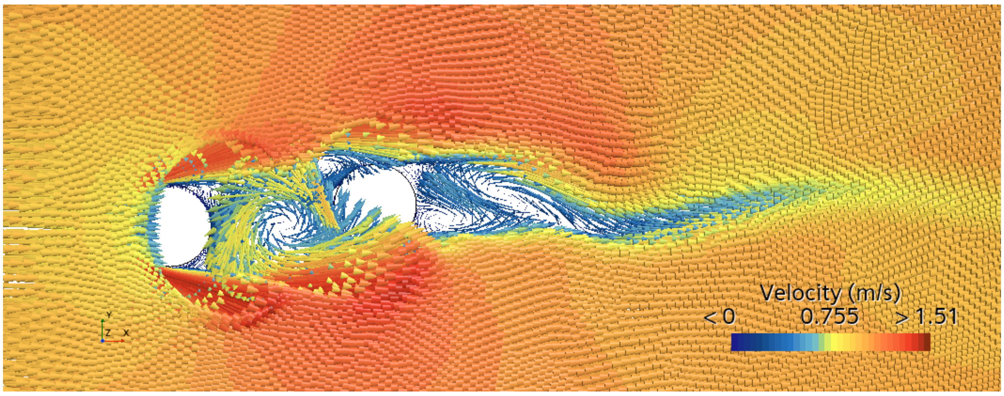


Figure 16: Velocity field for 1.5D cylinder using morpher at time = 10 [s]

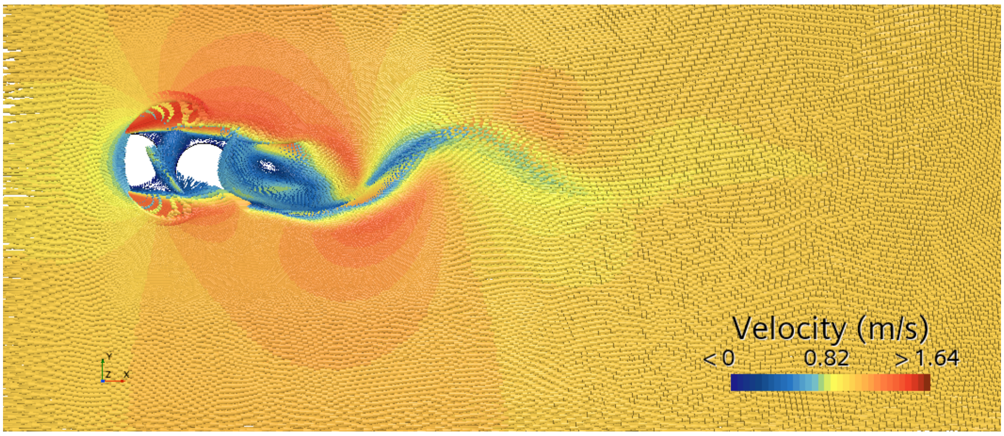


Figure 17: Velocity field for 0.25D cylinder using overset mesh at time = 2 [s]

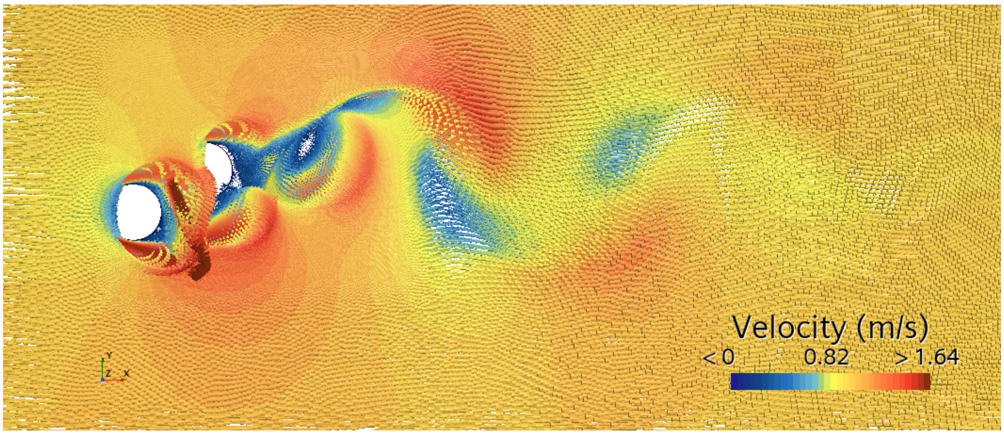


Figure 18: VVelocity field for 0.25D cylinder using overset mesh at time = 12 [s]

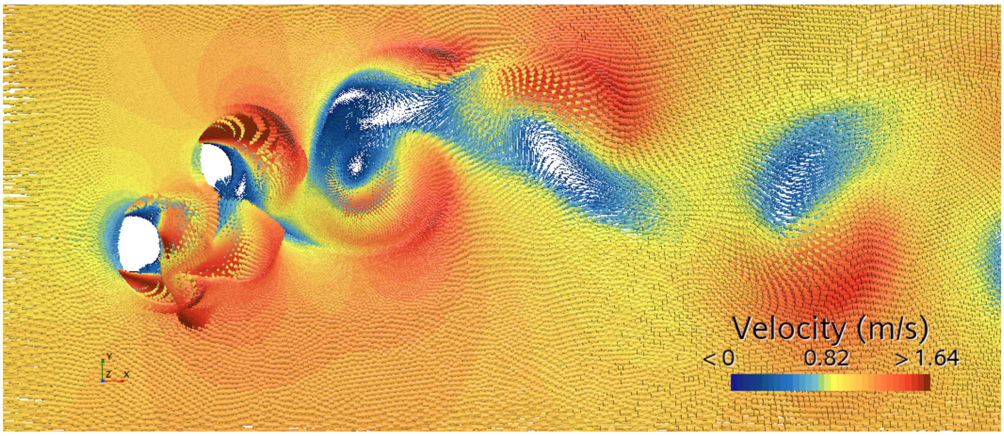


Figure 19: Velocity field for 0.25D cylinder using overset mesh at time = 15 [s]